

Depth-Informed Visual Navigation Policies for Goal-Conditioned Trajectory Generation

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Introduction

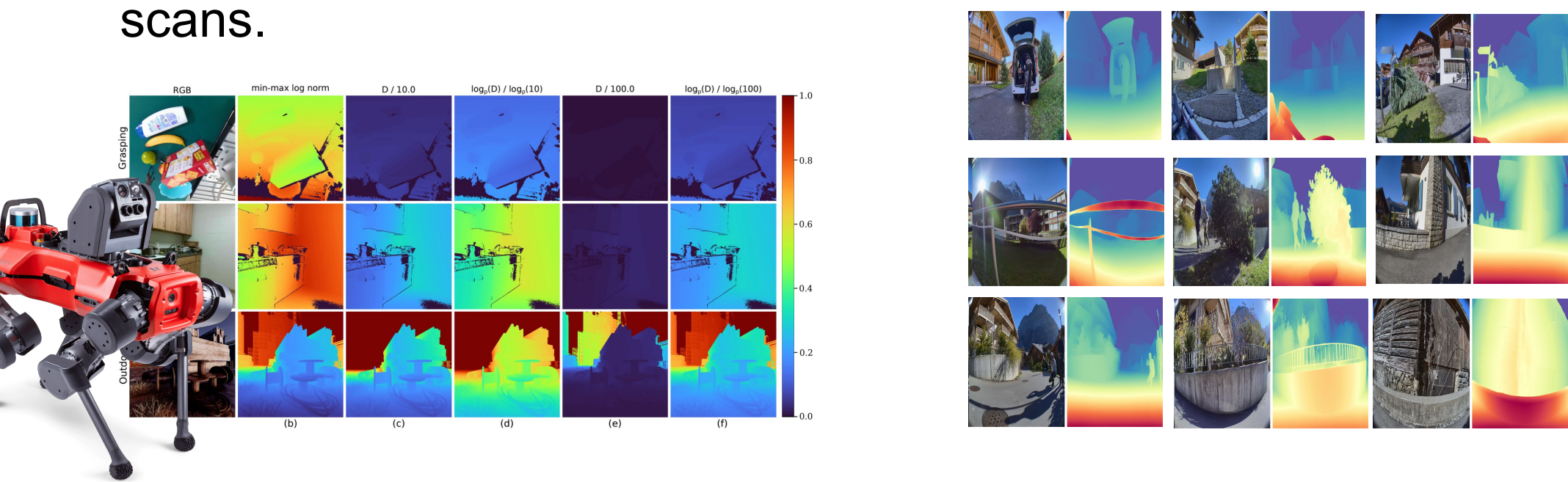
- In this project, we explore if **depth as a feature for goal-conditioned navigation** (conditioning on goal image and pose) enables more accurate trajectory prediction and faster generalization.
- First **(1)**, we implement and train an imitation learning baseline, using offline data from the four-legged robot data (front-camera). We also implement IL and PPO in Habitat for online data.
- Second **(2)**, we implement IQL-style AWR training to explore nearby plausible trajectory chunks for goal progress, smoothness, and success.
- Third **(3)**, we implement a MOPO-style algorithm that trains a dynamics ensemble over IL latent states and trajectory actions, rolls out synthetic transactions, and penalizes uncertain predictions (SAC-style actor/critic losses).

Background and Motivation

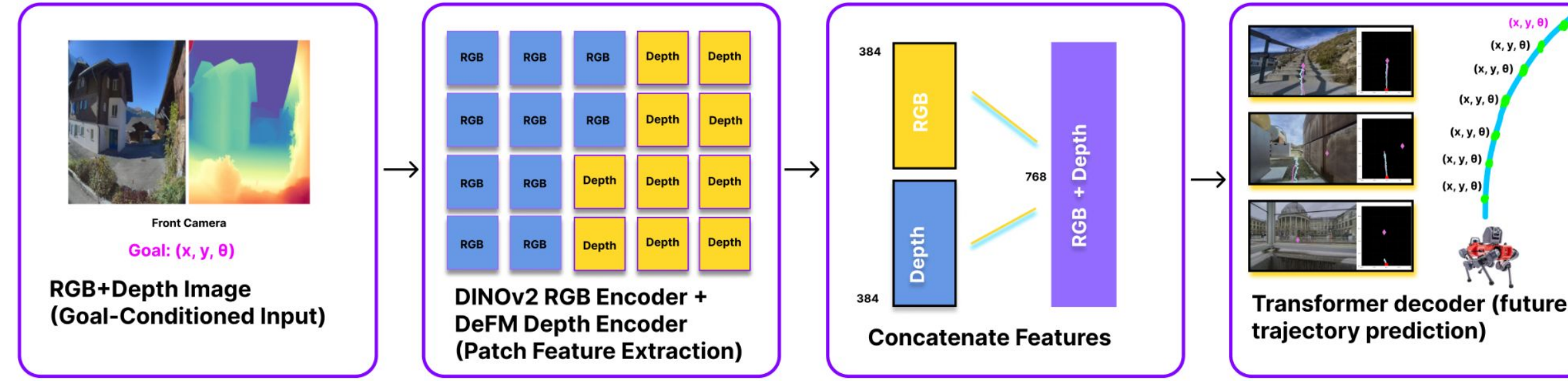
- Literature on goal-conditioned navigation with visual policies emphasize that conditioning on RGB images can be semantically sufficient, yet struggle with edge scenarios, where depth can be helpful. LiDAR maps are much more expensive, at times not meaningful for solving edge cases (e.g., objects).
- We build upon prior works such as LiMO and GrandTour to explore how geometric and spatial features, coupled with RGB semantics, can improve traversability cues, goal-distance estimation, and obstacle navigation (LiMO, GrandTour).
- We experiment with IL, IQL, PPO, and MOPO-style methods to understand if depth as a relevant feature holds in various navigation policies. We motivate the support of RGB semantics with depth-based geometric and spatial insight to understand how depth can support faster generalization and robust trajectory prediction.

Set-Up: Data

- We leverage the ANYmal GrandTour dataset (ETH) and the Habitat-Matterport 3D Research Dataset (HM3D).
- GrandTour consists of missions with 49 environments (indoor, outdoor, night, smoke, sand, urban, etc.) and is leveraged in offline algorithms.
- Habitat is leveraged for IL and PPO (off-policy, online) experimentation. HM3D consists of 1,000 high-resolution 3D scans.



Methodology

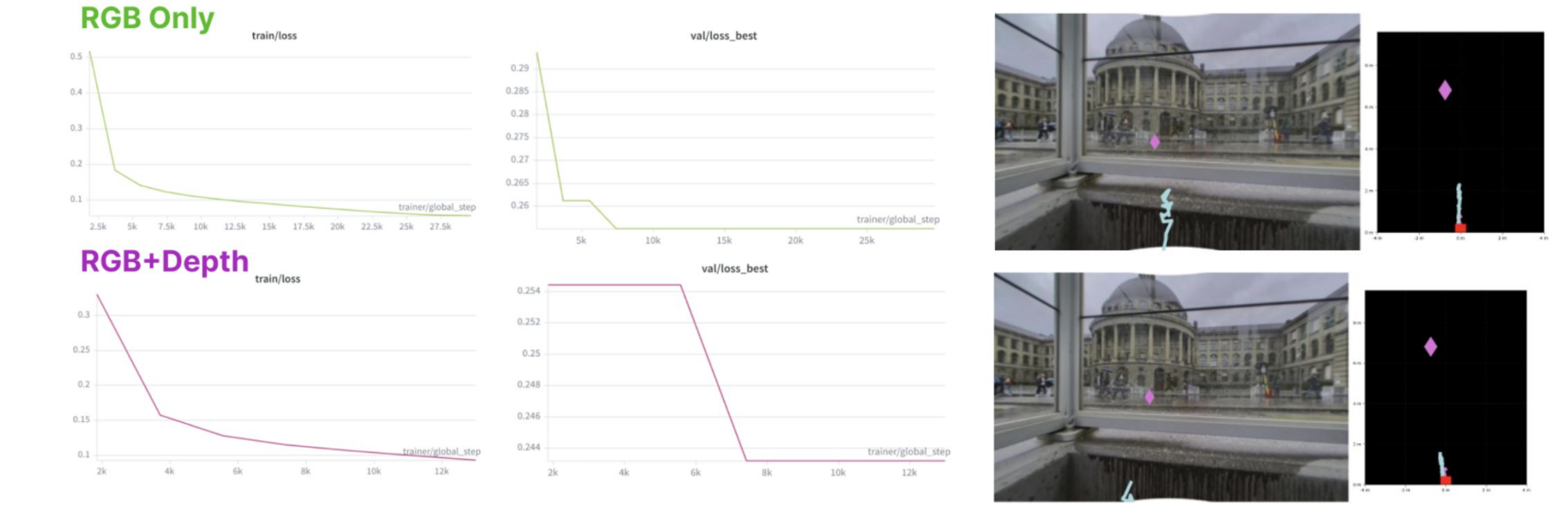


Algorithm	Learning Signal	Objective	Key Idea
BC (IL)	Expert trajectories	$\mathcal{L}_{BC} = \ \hat{\tau} - \tau^*\ _2^2$	Supervised waypoint regression that directly imitates expert trajectories.
AWR	Offline rewards + demonstrations	$\mathcal{L} = \mathcal{L}_\pi + c_V \mathcal{L}_V$	Offline RL with expetile value estimation and advantage-weighted imitation.
MOPO	Offline transitions	$\tilde{r} = r - \lambda u(s, a)$	Model-based offline RL that penalizes uncertain model rollouts using ensemble disagreement.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

$$\text{SPL} = \frac{1}{M} \sum_{i=1}^M S_i \frac{\ell_i}{\max(p_i, \ell_i)}$$

Analysis



- Policies augmented with real depth achieve lower validation loss than RGB-only policies.
- RGB+Depth achieves best performance earlier in training (suggesting geometric information improves sample efficiency and accelerates generalization)
- RGB+Depth achieves smoother paths and improved obstacle avoidance when depth is available (depth provides complementary spatial cues).

Key Takeaways

- Depth improves navigation performance in both offline imitation learning and online RL settings (GrandTour, Habitat).
- Depth potentially accelerates learning depending on the task, reaching strong performance metrics within fewer training steps.
- Geometric information, not necessarily additional input channels, enable depth-influenced reasoning.
- Depth offers similar success rates, but higher SPL an lower distance to goal for trajectories.

Future Work & Limitations

- Scale experiments to 100+ missions on even more diverse scenarios that may entail object-handling, grasping, etc.
- Vary mediums of depth quality for features and signals (planned).
- Extend policy with a VLA model and investigate whether language instructions improve performance in long-tail navigation (planned).
- Assess sim-to-real transfer with real-world robot deployment and with a cross-embodiment dataset (for e.g., for grasping, etc.)
- Modality-aware policy that dynamically weighs RGB and depth features on environmental uncertainty.

References

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Results - Experiments

Table 1: Comparison of RGB and Depth-Based Navigation Models

Model	Val Metric @ 7k/Final	Best Metric	Steps to Best
RGB Only (IL)	0.259	0.255	~27k
RGB + Real Depth (IL)	0.243	0.243	~7k
RGB + Black Depth (IL)	0.255	0.246	~2k
RGB Only (AWR)	0.296	0.295	~4.5k
RGB + Depth (AWR)	0.286	0.285	~6k
RGB Only (MOPO)	3.03	2.96	~1k
RGB + Real Depth (MOPO)	3.13	3.09	~3k

